

Handwritten Mathematical Problem Solver

IPCV

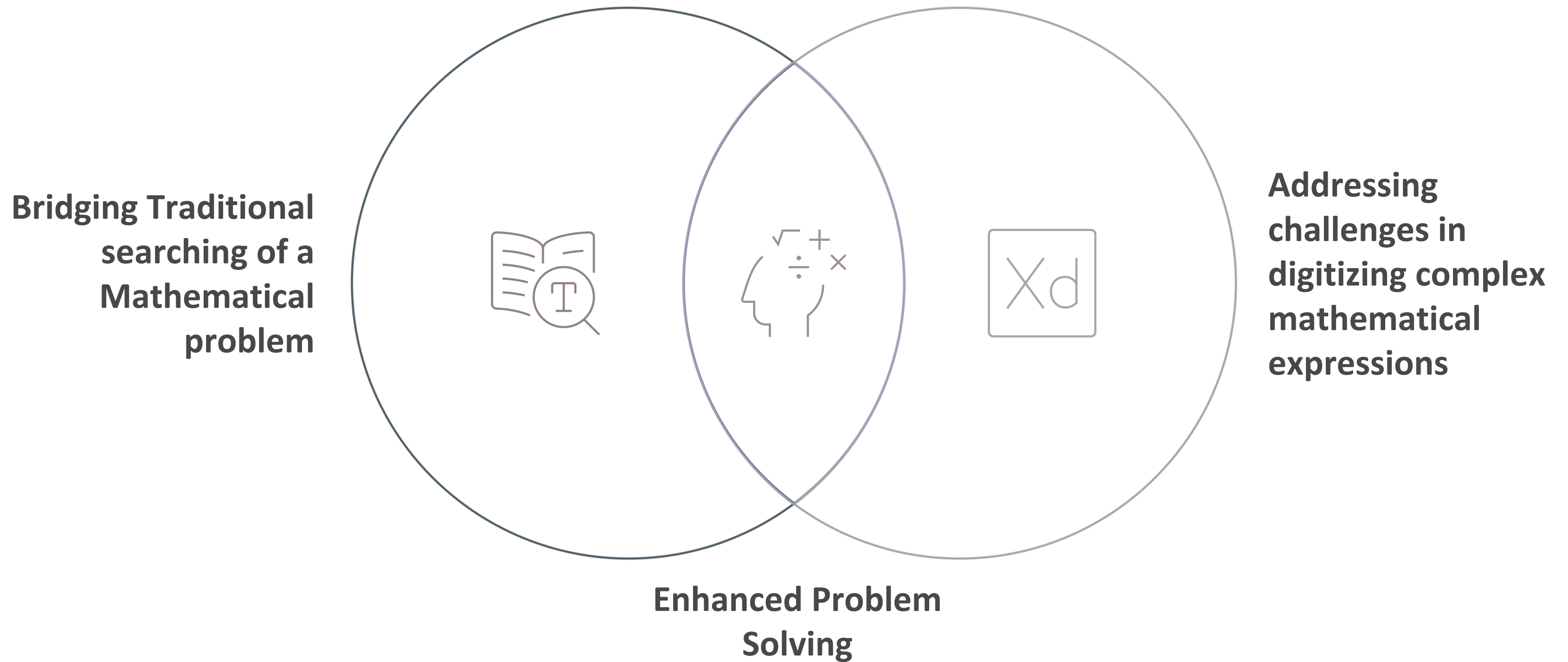
**Sandip
Katel**

**Saphal
Rimal**

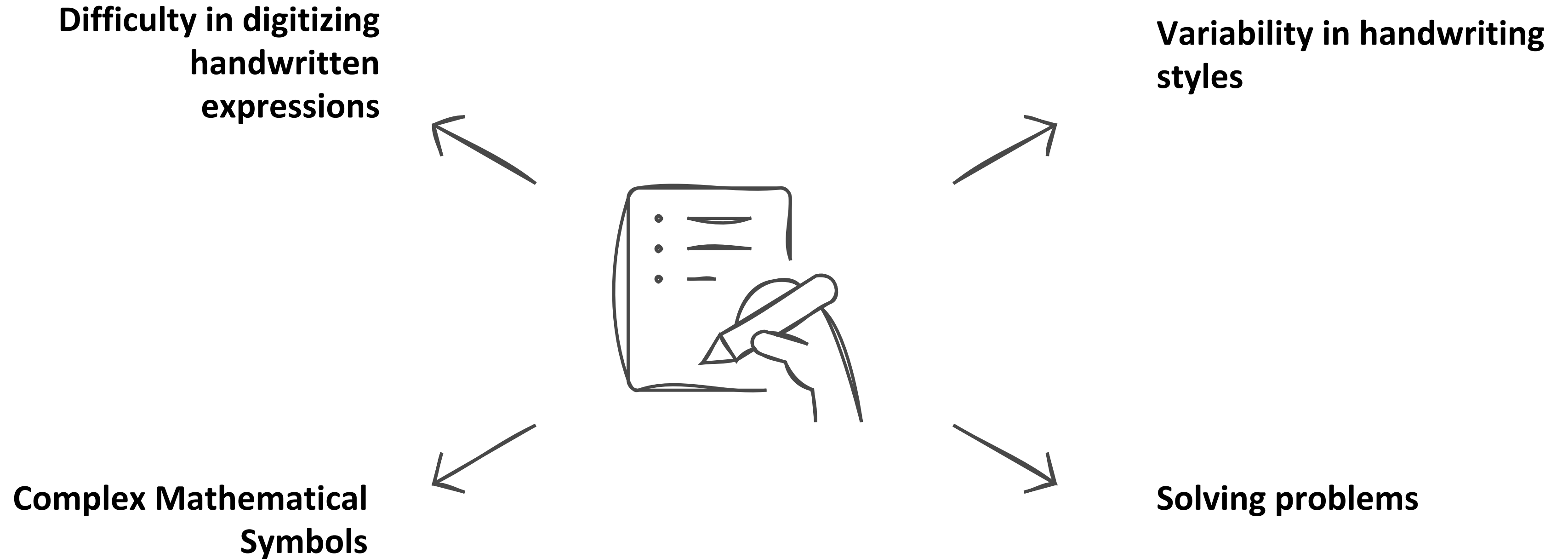
**Sharad
Pokharel**

**Sijan
Joshi**

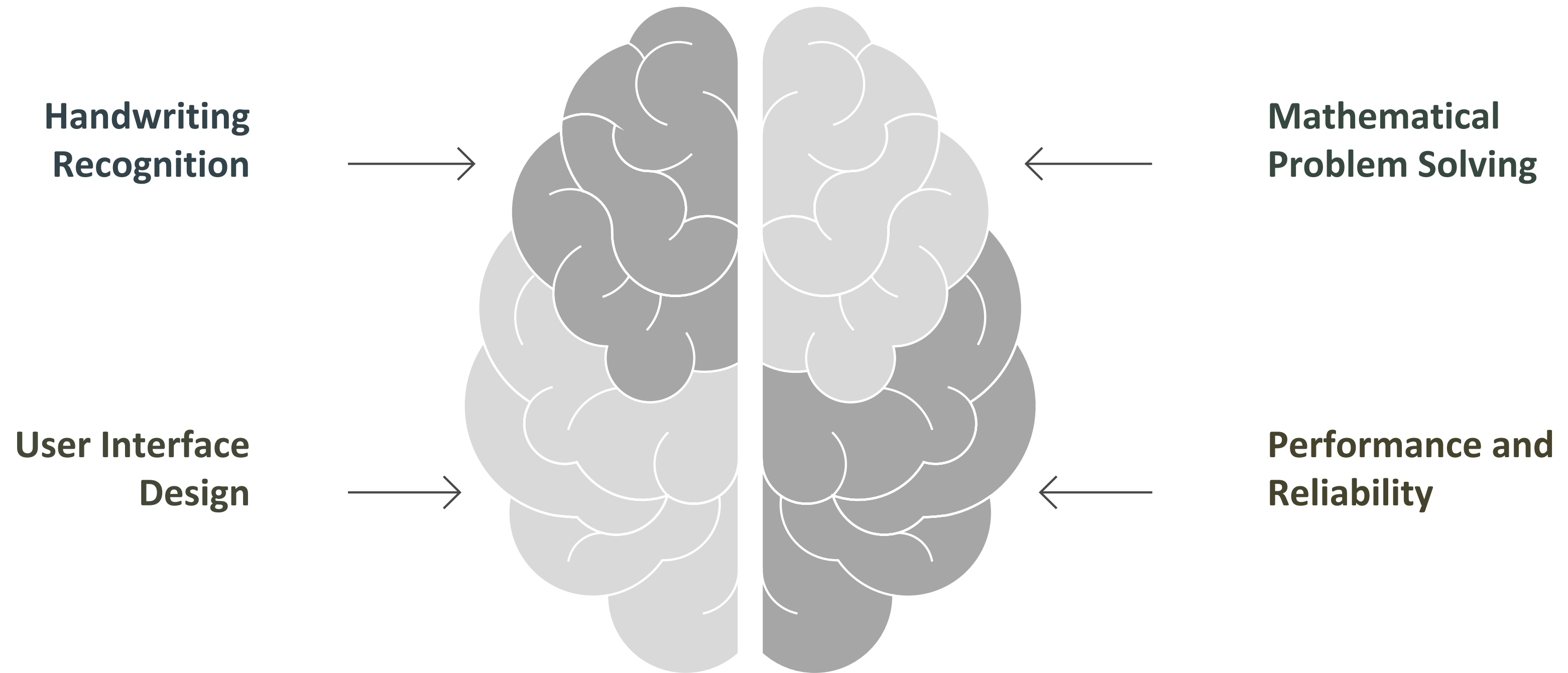
Project Overview: Introduction



Project Overview: Problem Statement



Project Overview: Objectives



Related Works

Socratic

Struggles with complex notations and spatial arrangements



Mathway

Limitations with cursive handwriting and complex layouts



Photomath

Sensitive to image quality and messy handwriting



Gauthmath

Slower response times due to human verification

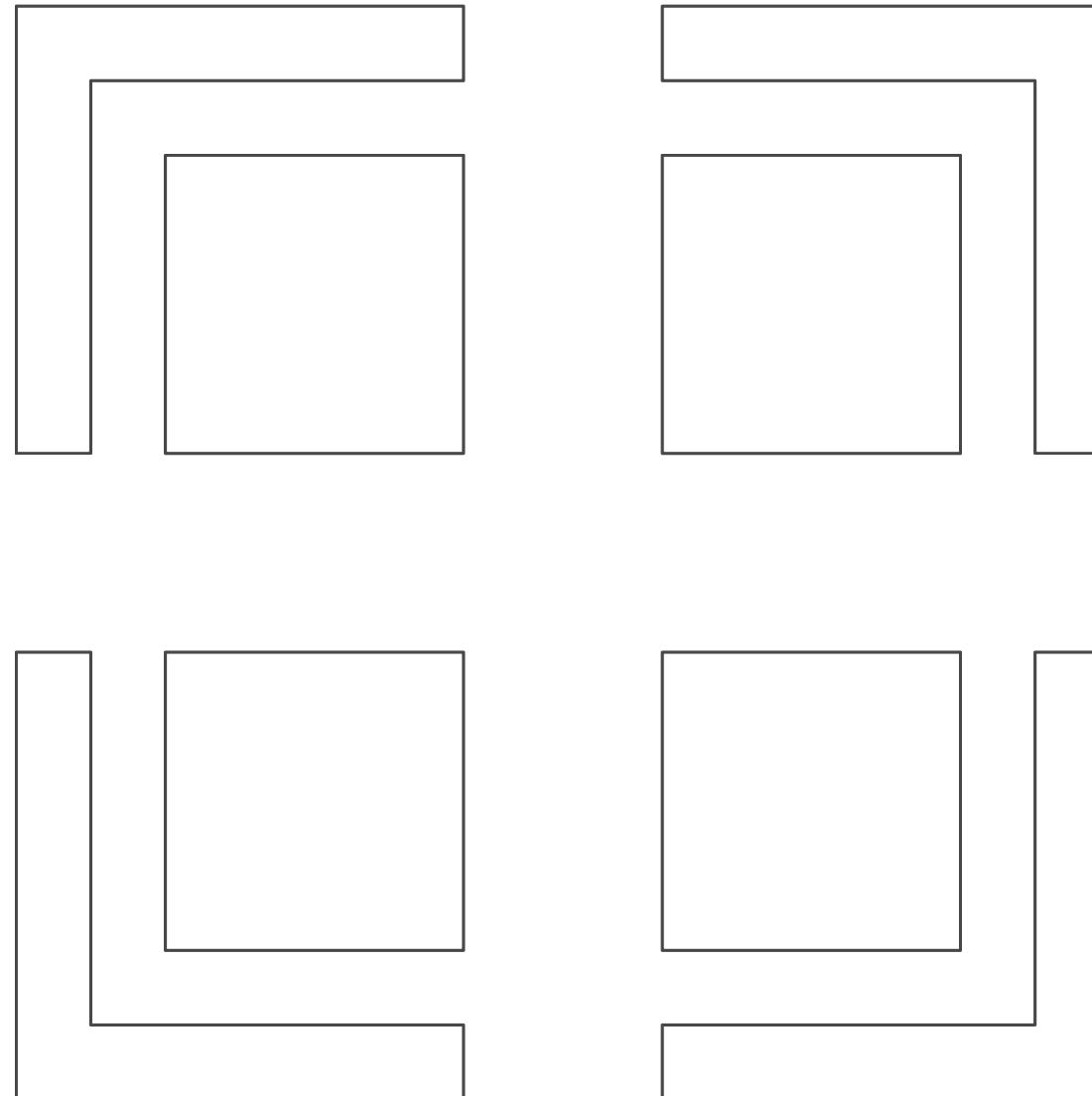


Related Theory

Solving

Computer Algebra Systems

SymPy for symbolic computation and expression manipulation.

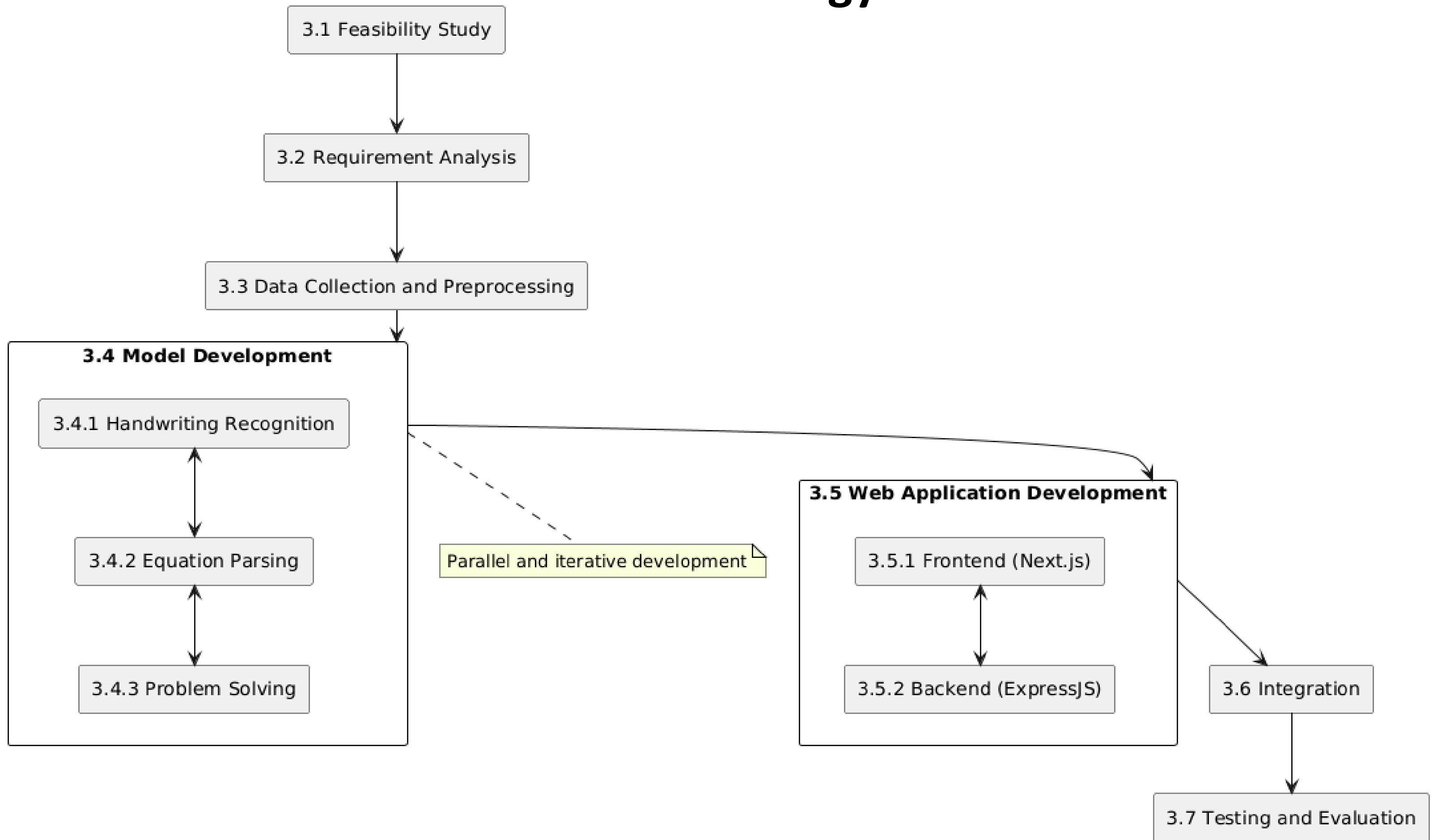


Detecting

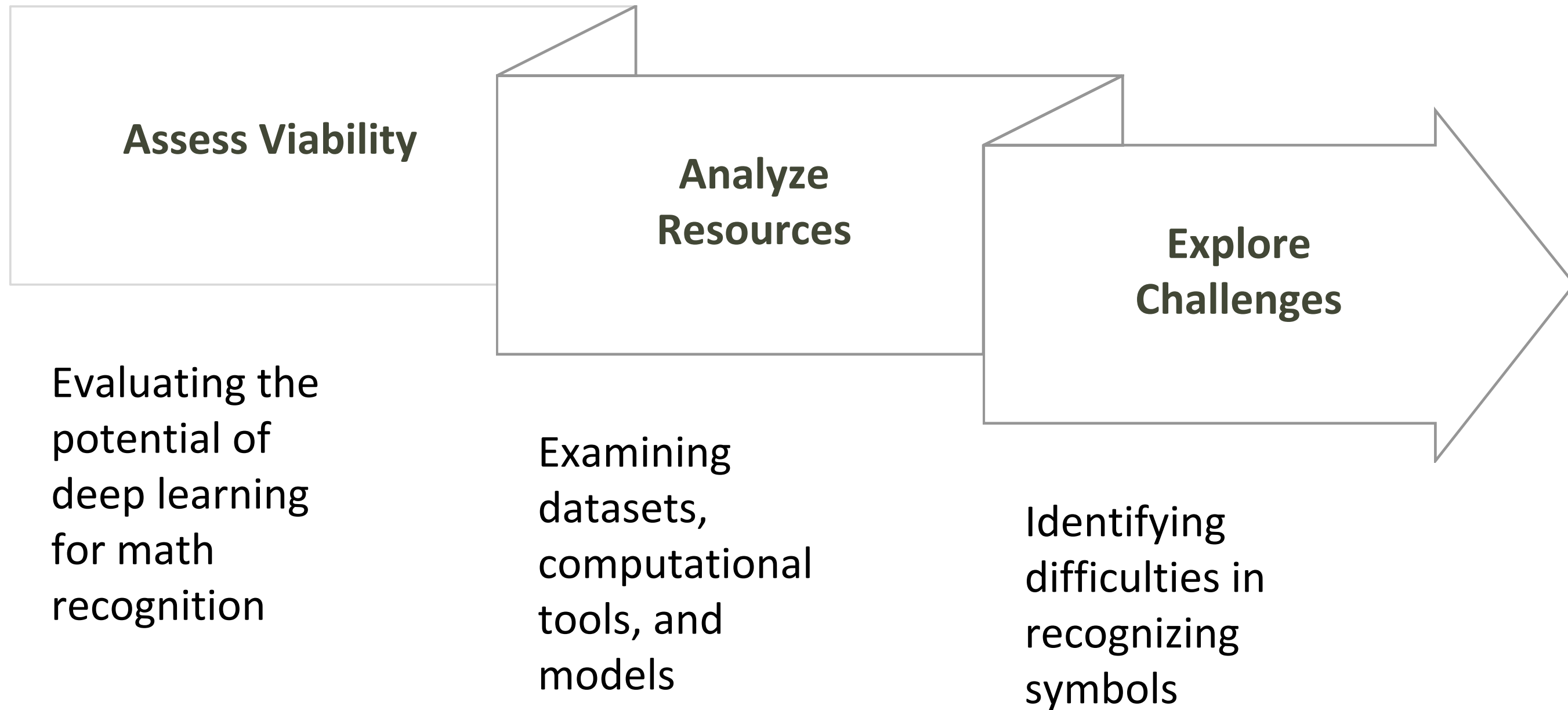
TrOCR Framework

A transformer-based OCR system capturing long-range dependencies and context.

Methodology



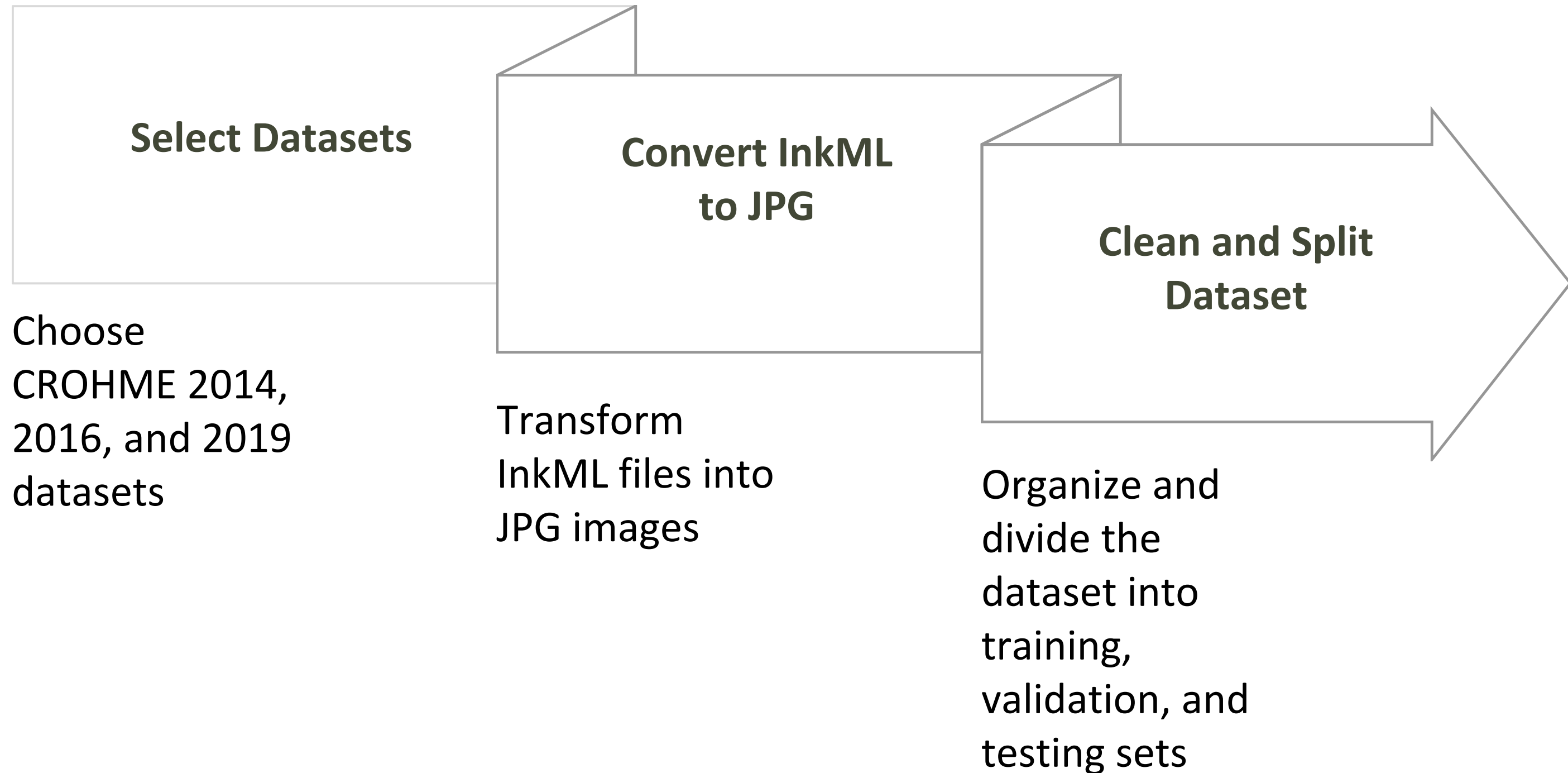
Methodology: Feasibility Study



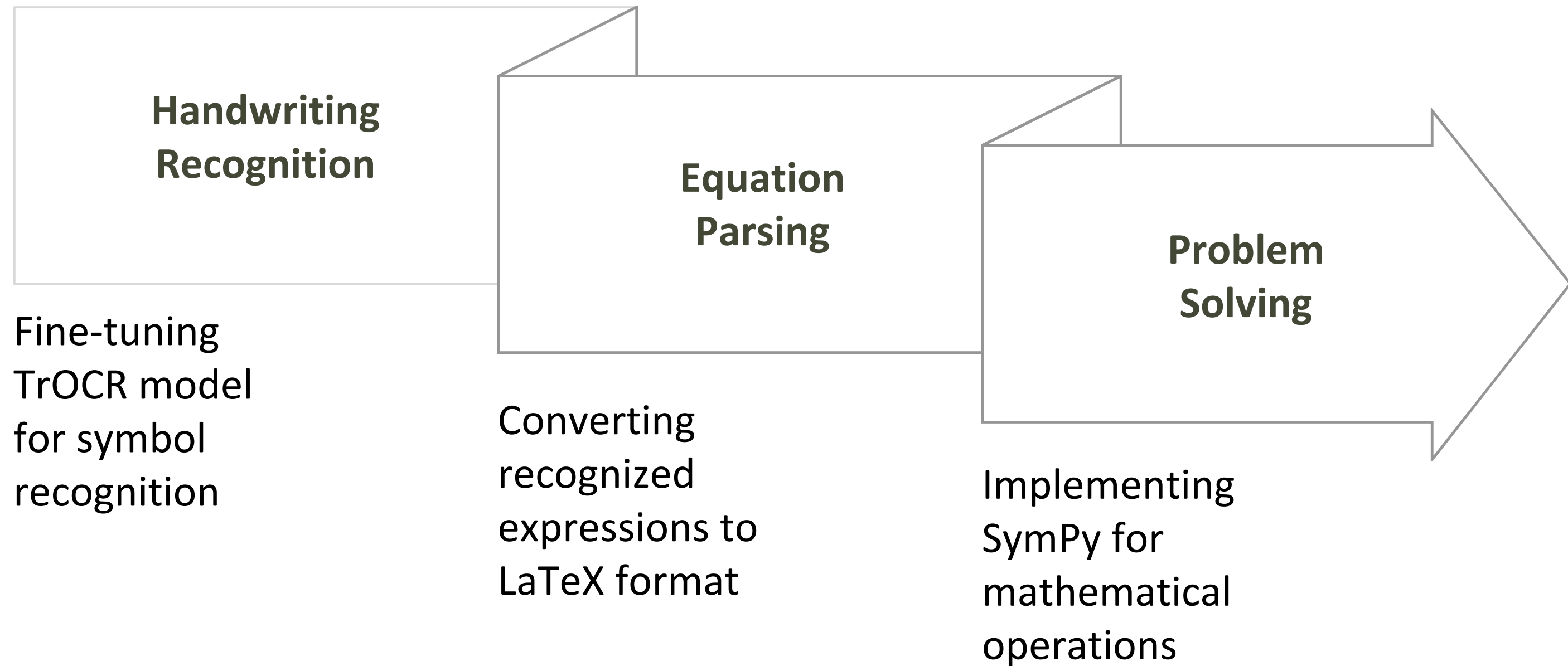
Methodology: Requirement Analysis

Requirement	Functional	Non-Functional
Recognition	Expression Recognition	Model efficiency
Representation	Manipulate recognized expression	Usability
Processing	Equation solving	Performance

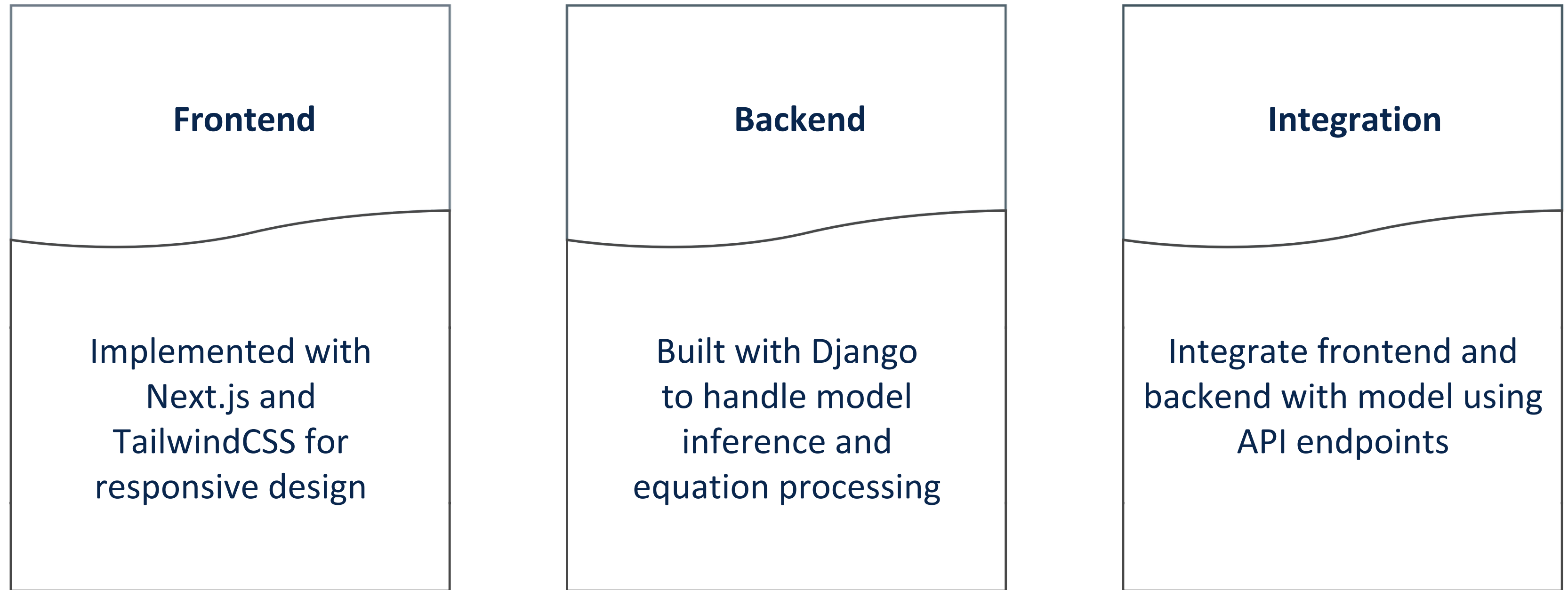
Methodology: Data Collection and Preprocessing



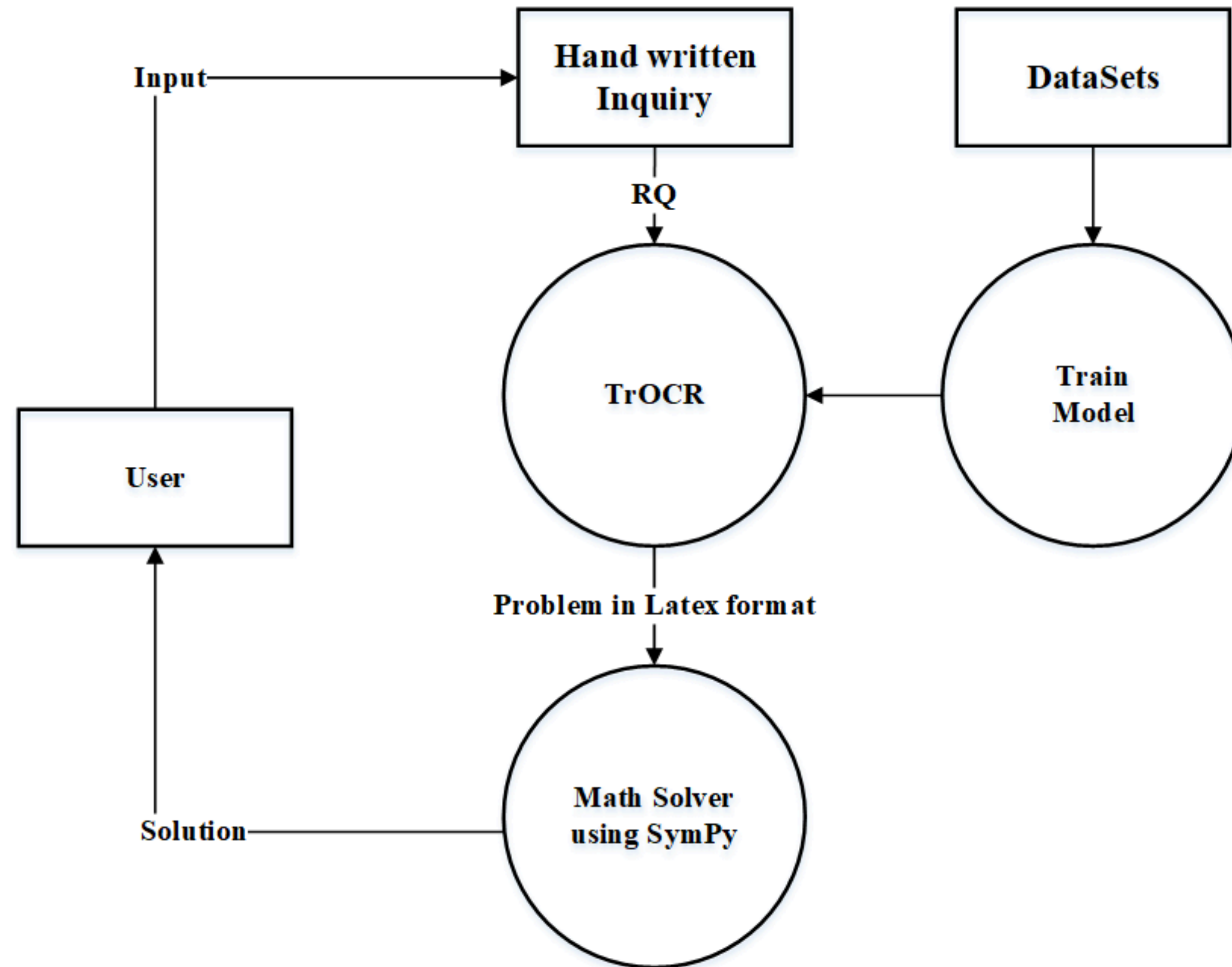
Methodology: Model Development



Methodology: Web Application Development



Implementation: Context Diagram

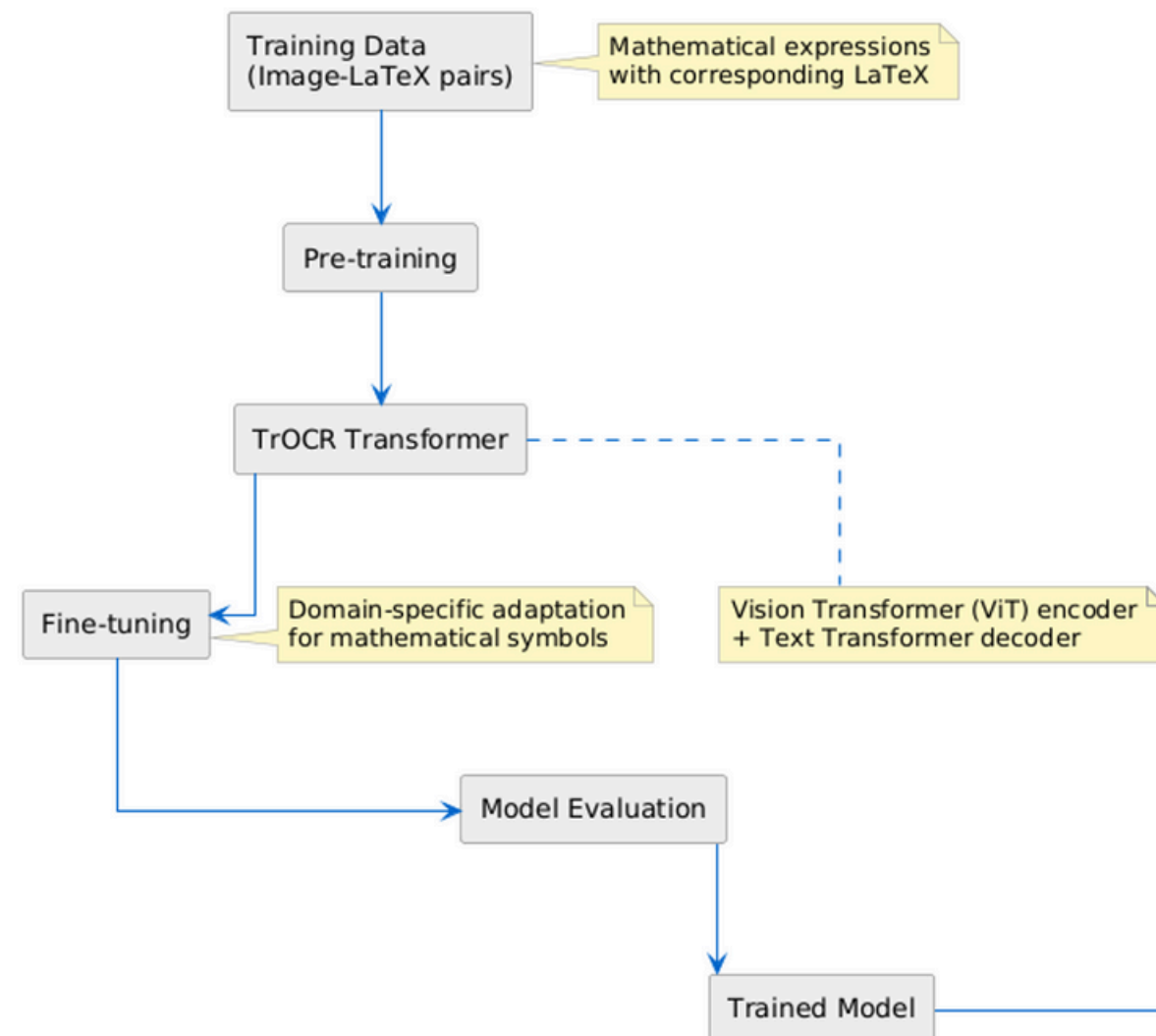


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System Architecture

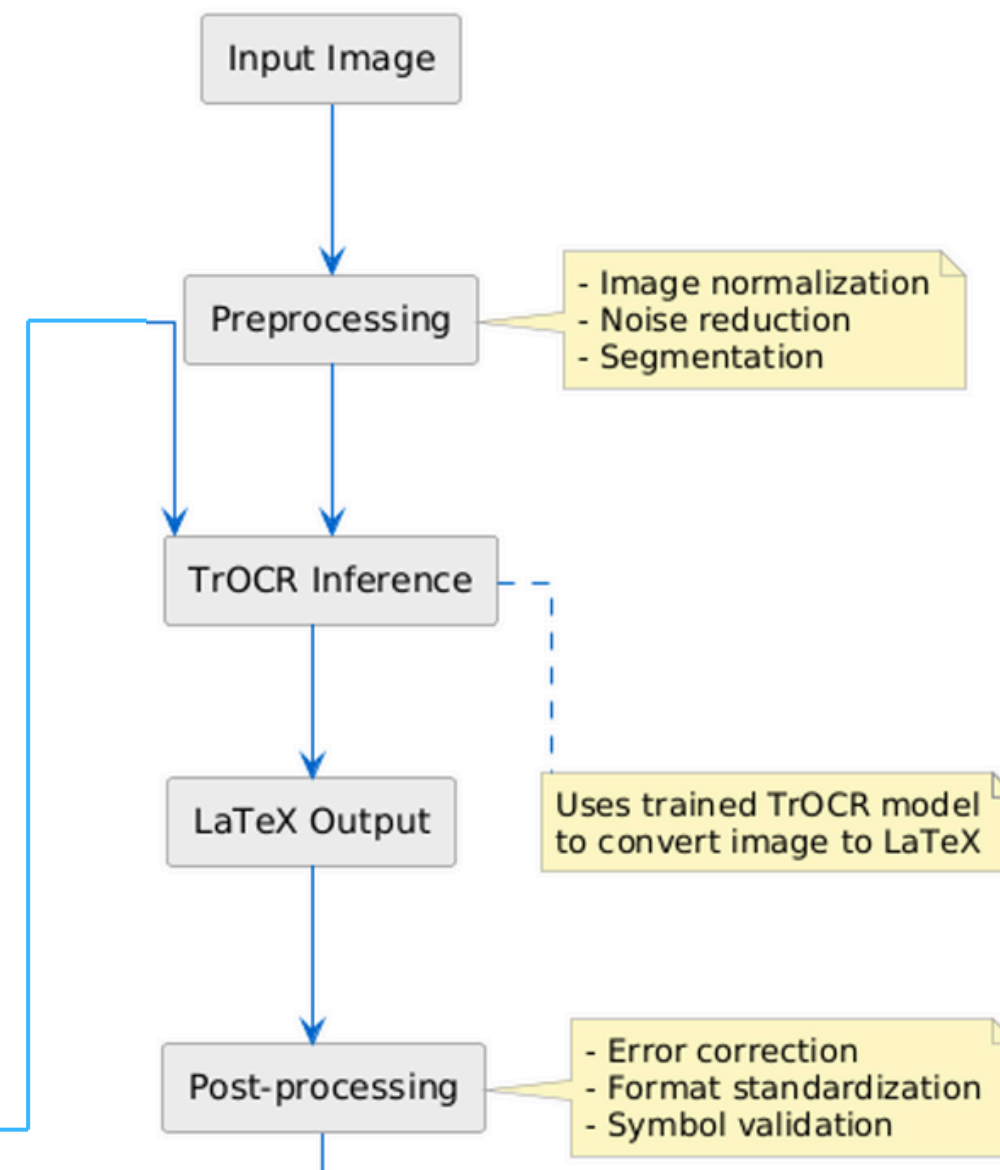
Model Training Module

Prepare TrOCR for mathematical expression recognition



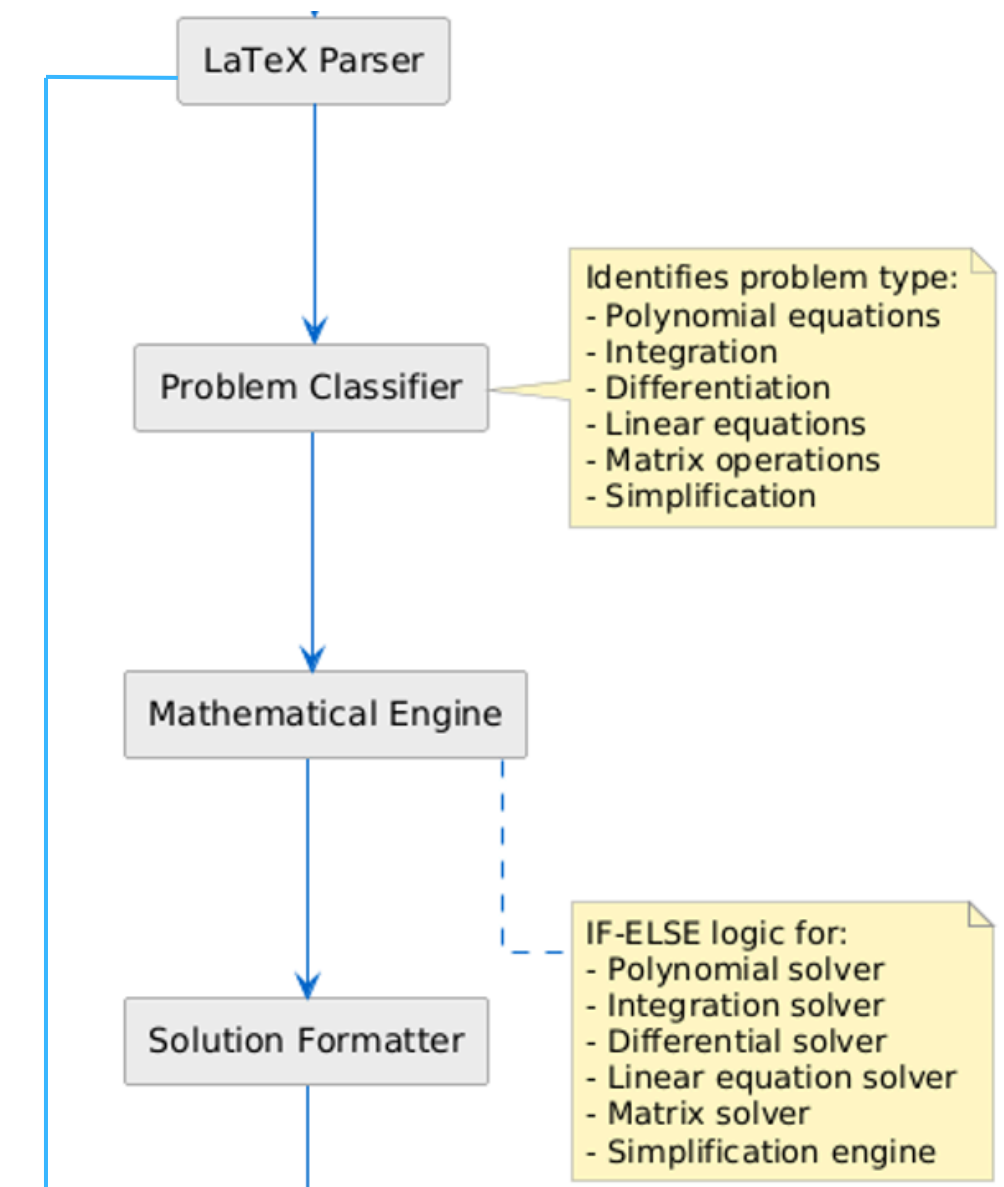
Character Recognition Module

Convert handwritten input to LaTeX

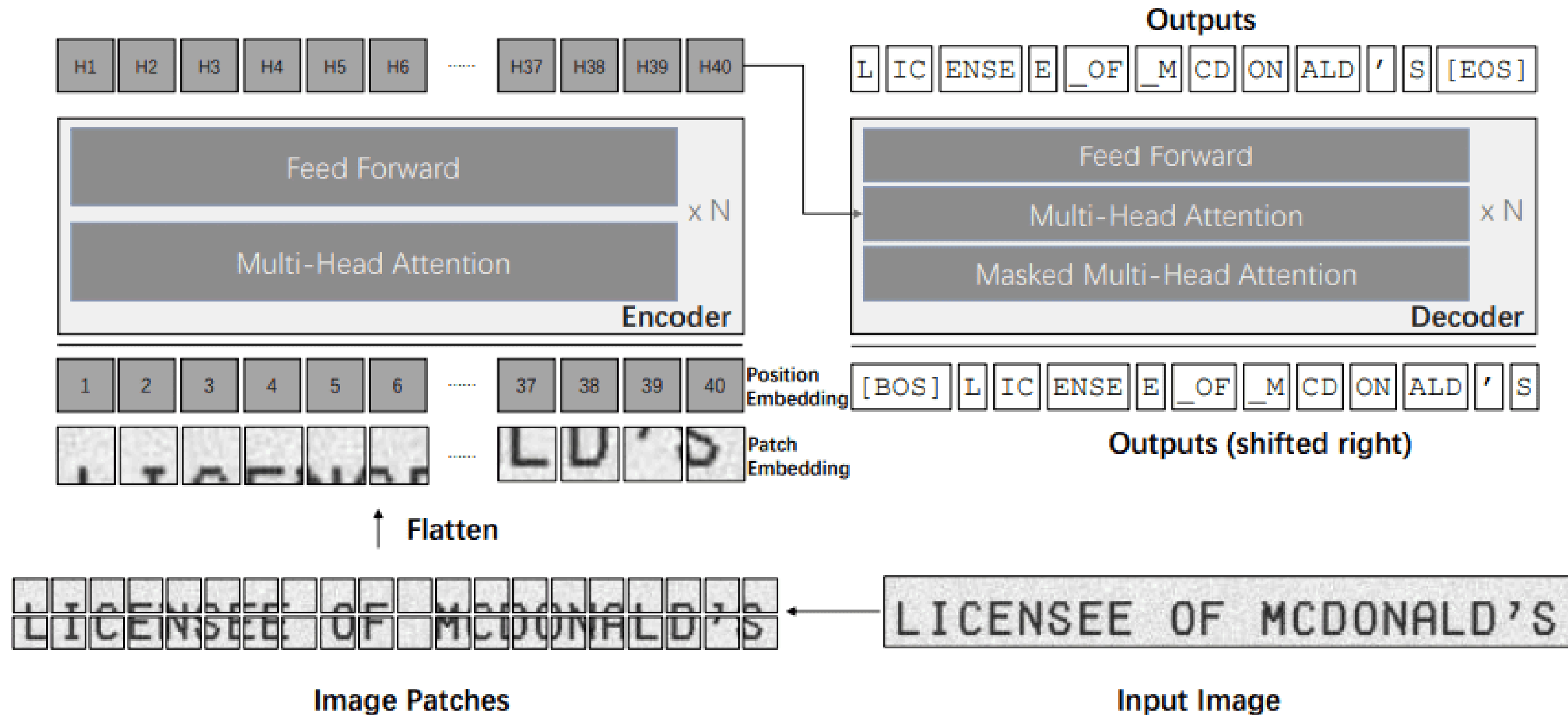


Solution Generation Module

Parse and solve mathematical problems

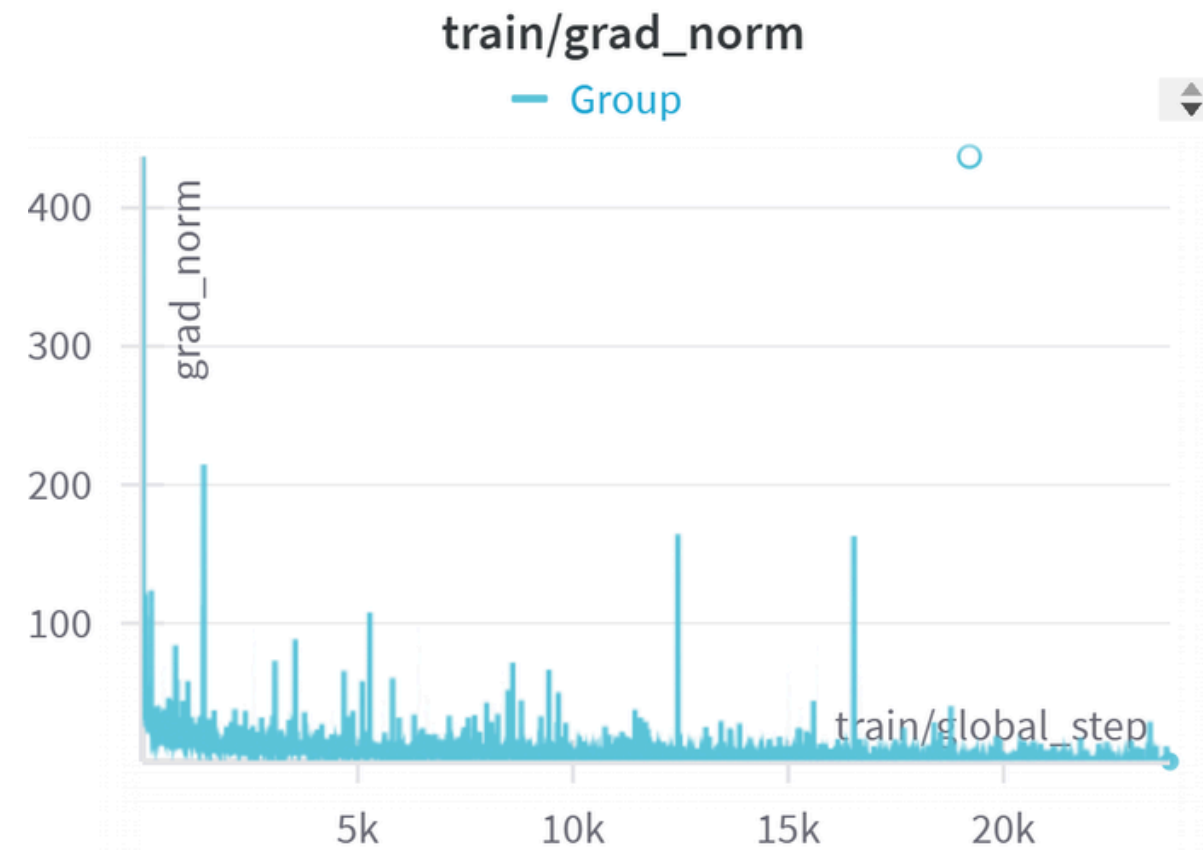
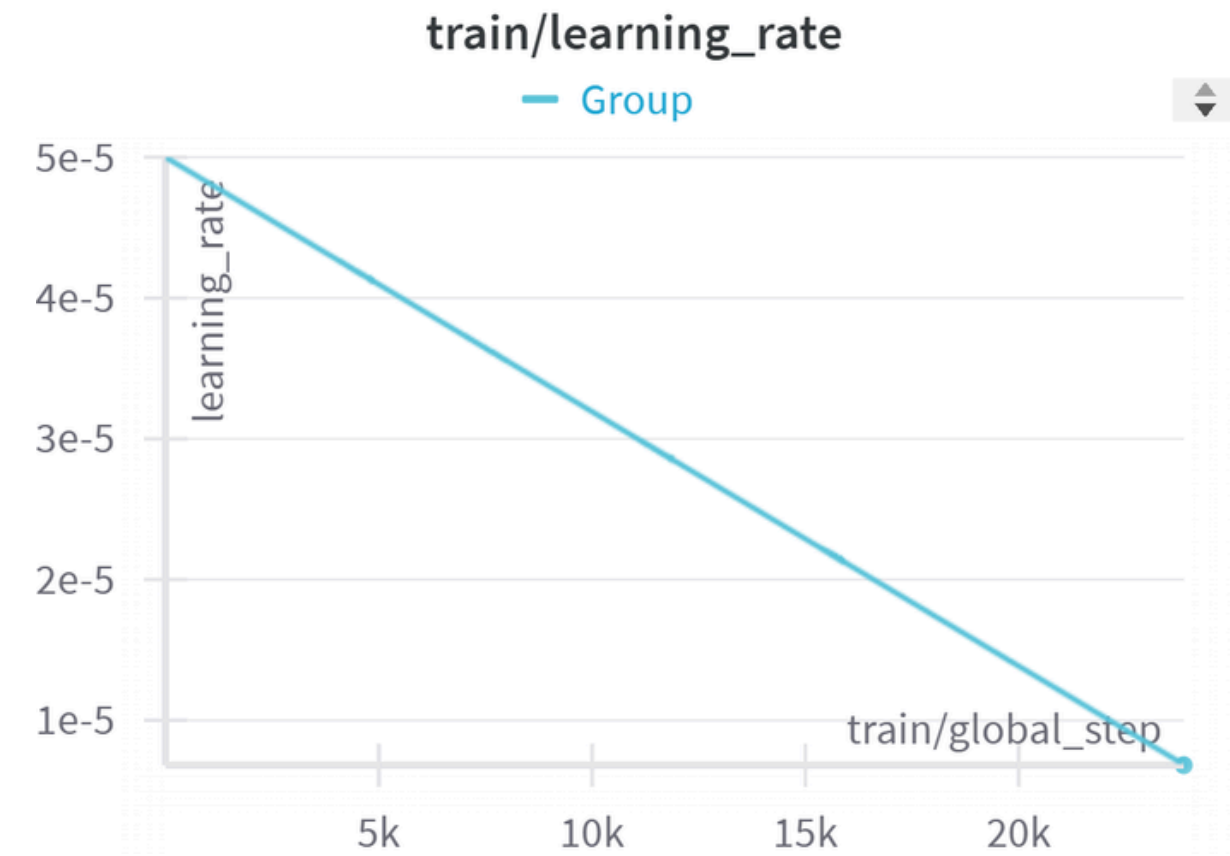
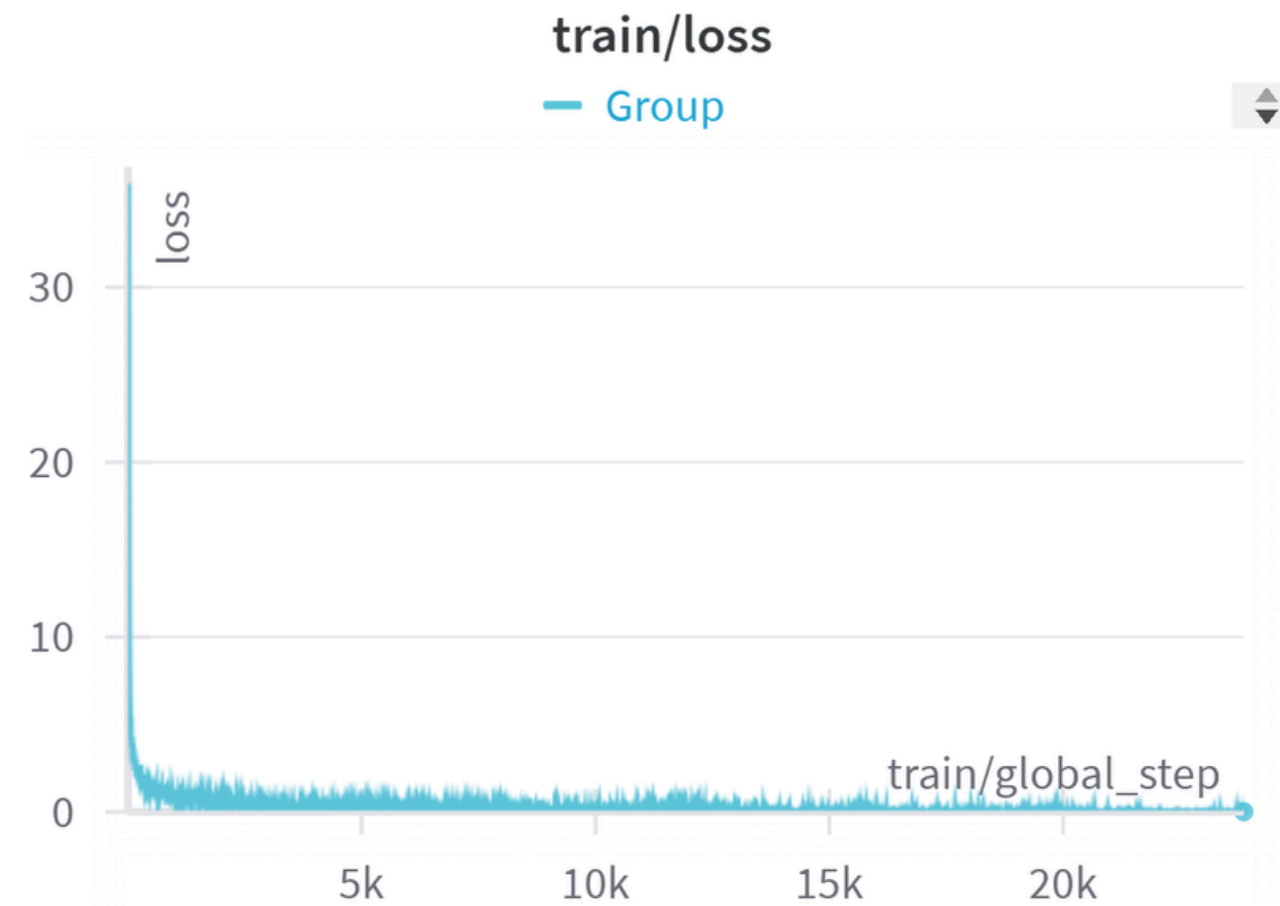


System Architecture: Transformer-based Optical Character Recognition (TrOCR)



TrOCR architecture. Taken from the [original paper](#).

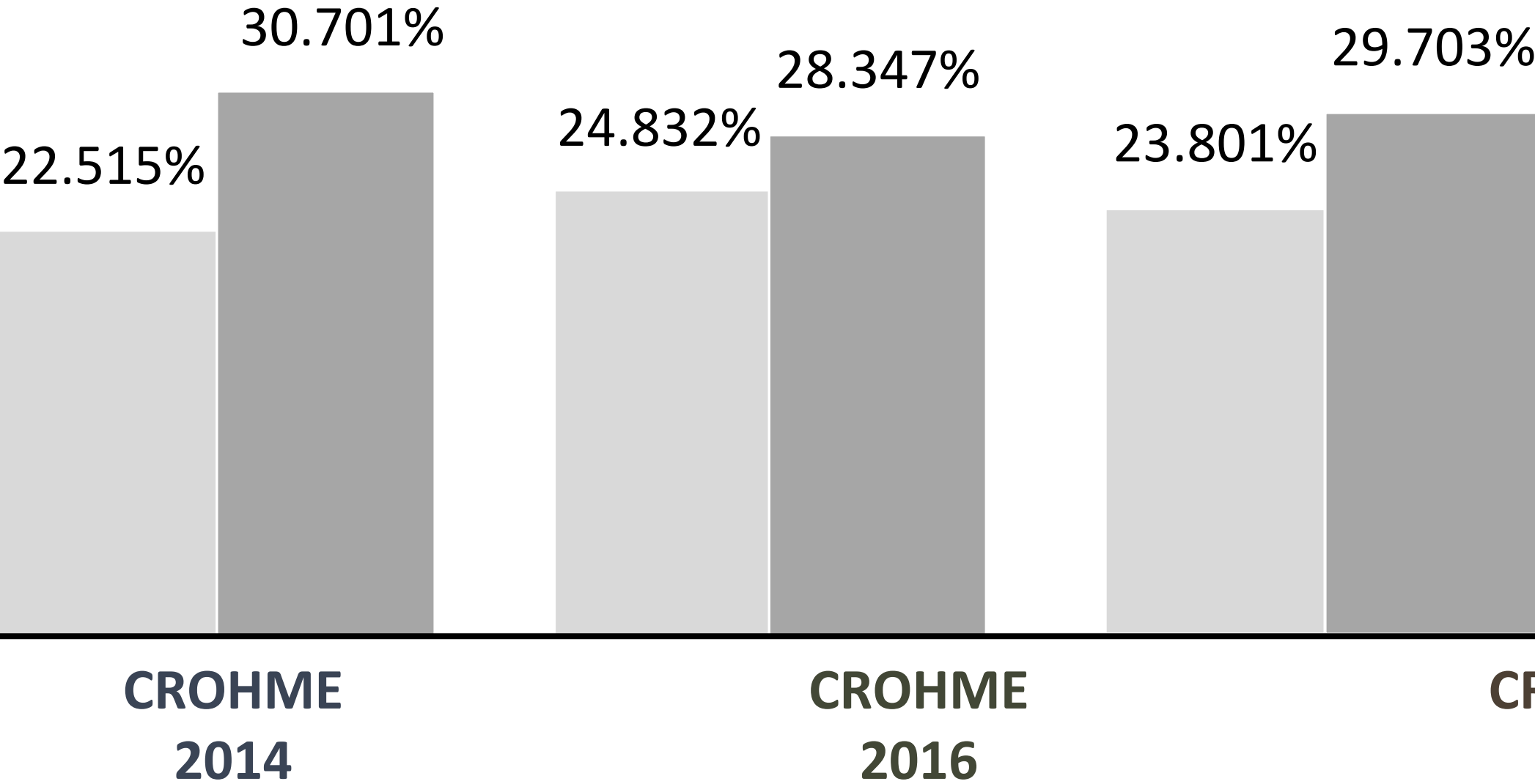
Result: Training Performance



Result: Test Results

Performance
Metric

Accuracy
Error Rate



Inference Time
15 seconds to 1 minute



Result: Validation Methods



Recognition Accuracy

Measured accuracy across multiple datasets.



Parsing Success

Assesses parsing success rate for equation formatting.



Solution Accuracy

Verifies solution accuracy against expected results.



Performance Monitoring

Continuously monitors model performance using Weights & Biases.

Result: Significance of Results



Bridging Handwritten and Digital Notation

Successfully connected handwritten notation with digital computation



Accessible Tool Creation

Developed a user-friendly tool for students, educators, and professionals



Effective Integration of TrOCR and SymPy

Demonstrated successful integration of TrOCR and SymPy for mathematical problem-solving



Foundation for Future Enhancements

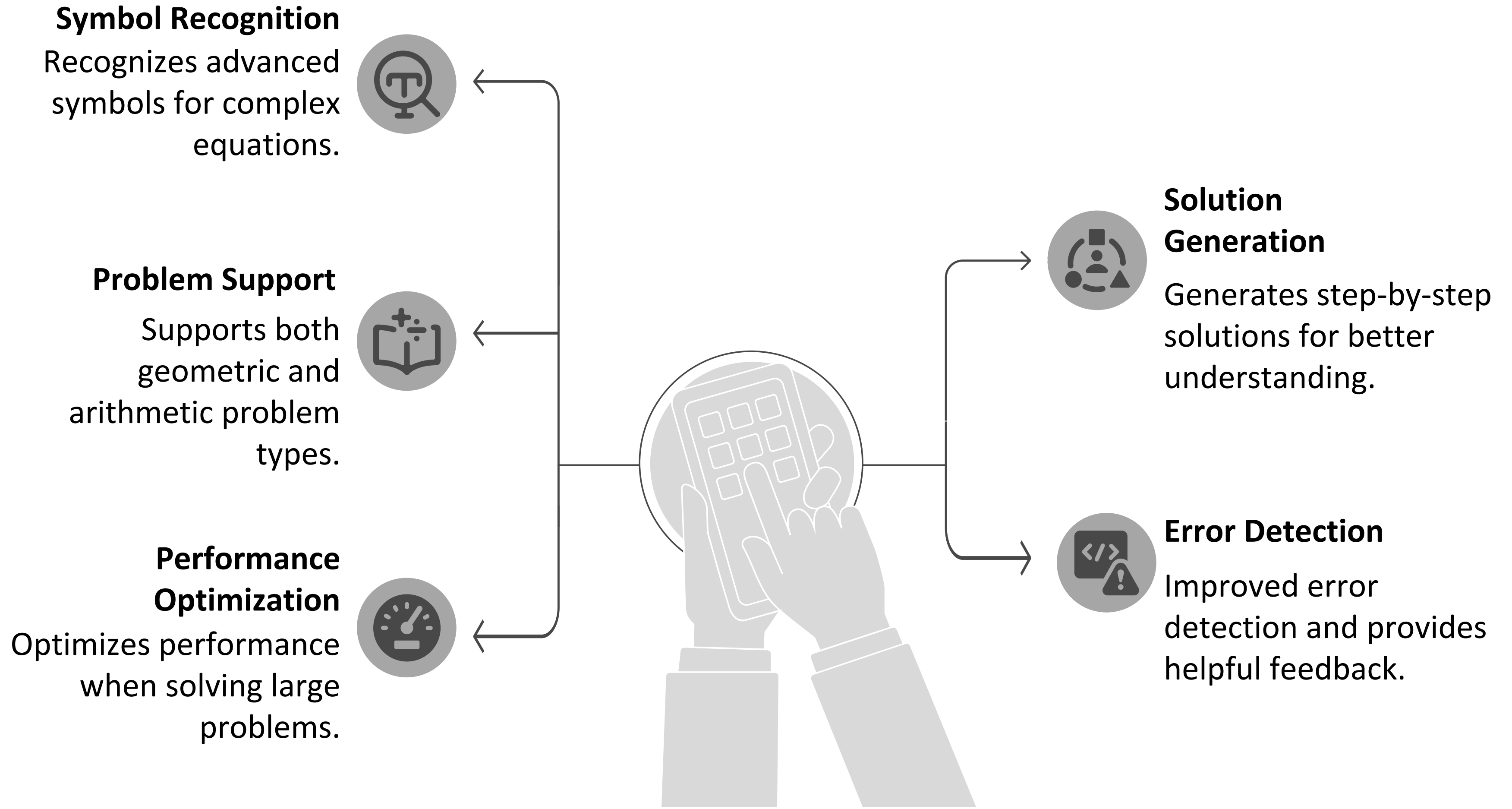
Established a base for future improvements in mathematical recognition systems

D E M O

Current Limitations

	Detection Section	Solving section
1	Limited recognition of complex mathematical symbols	Limited Scope of Solvable Problems
2	Dependency on legible handwriting and background	No step-by-step solution generation
3	Challenges with non-standard input formats	Absence of error checking for solution accuracy

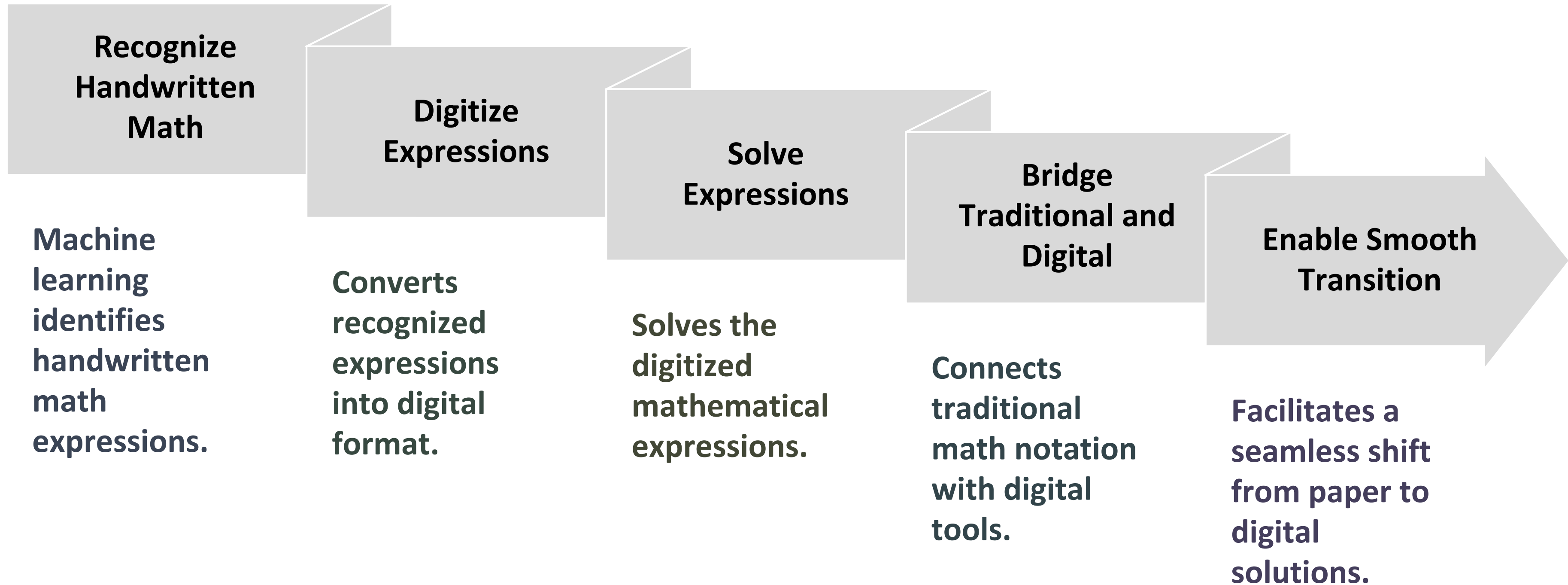
Future Enhancement

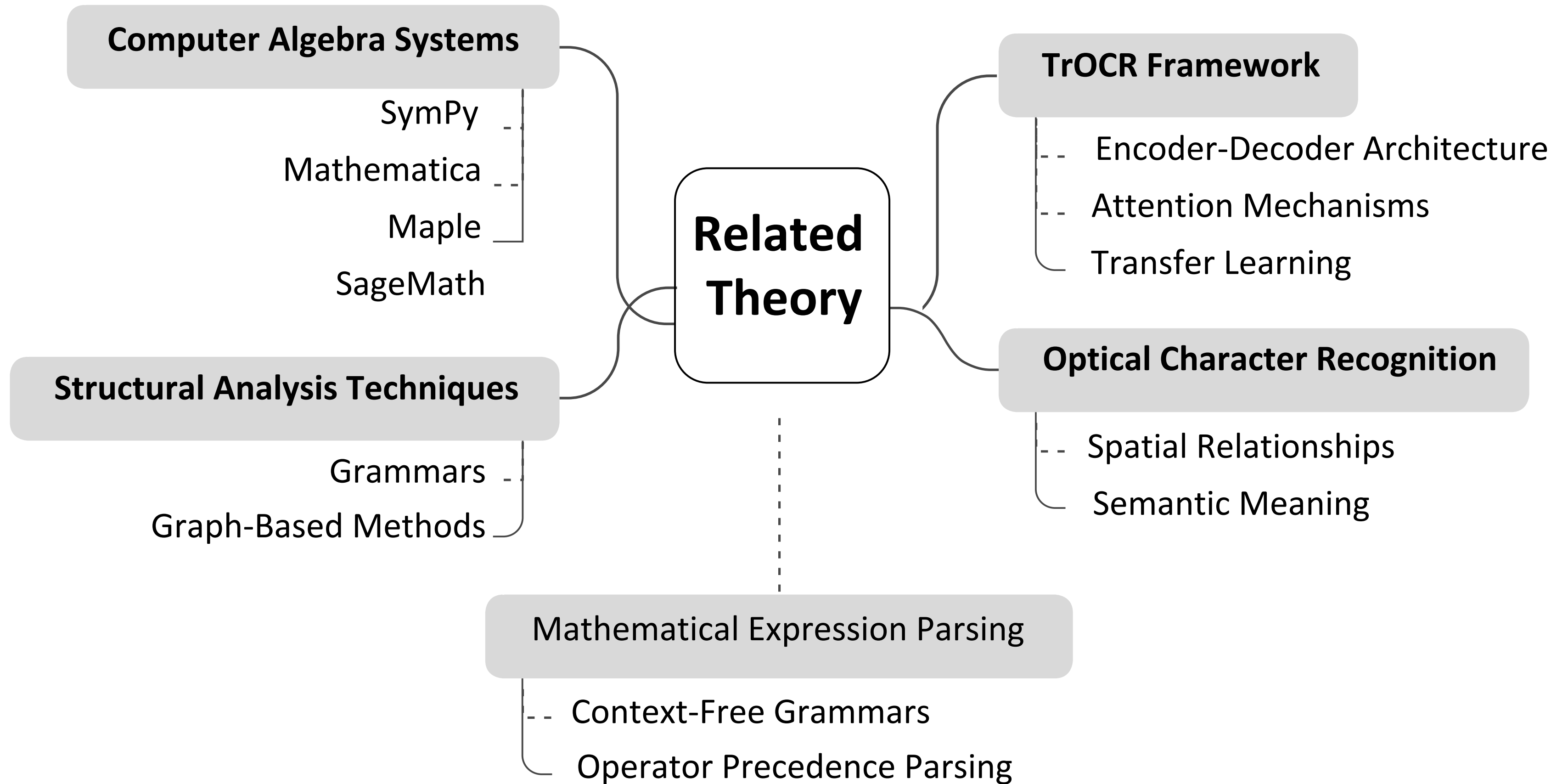


Thank You

Do you have any questions?

Project Overview: Introduction

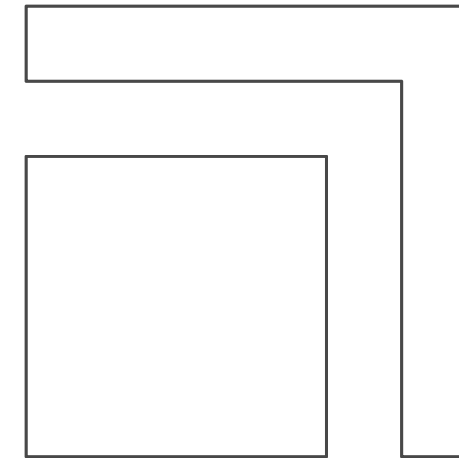
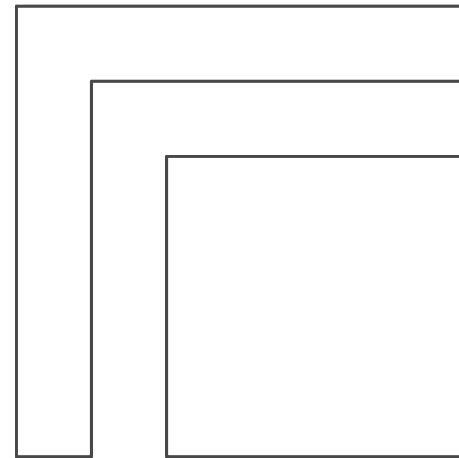




Related Theory

Computer Algebra Systems

Tools like SymPy for symbolic computation and expression manipulation.

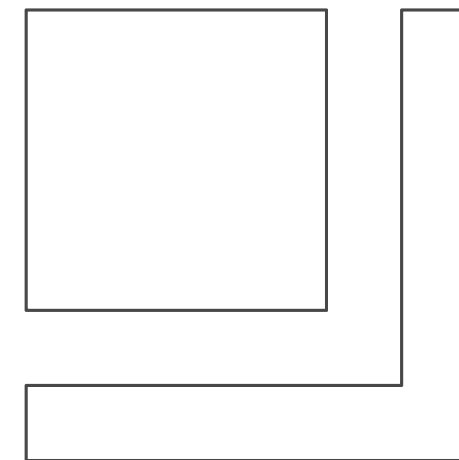
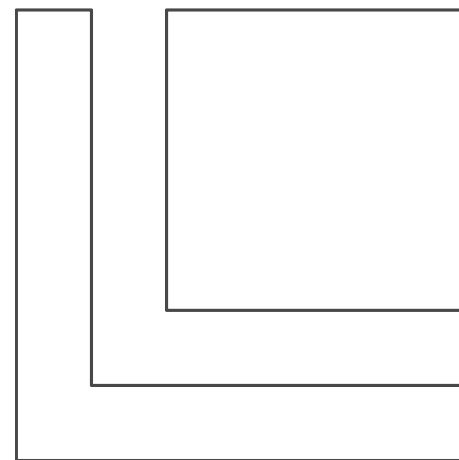


Structural Analysis

Techniques that decompose expressions into elements and analyze spatial relationships.

Parsing Techniques

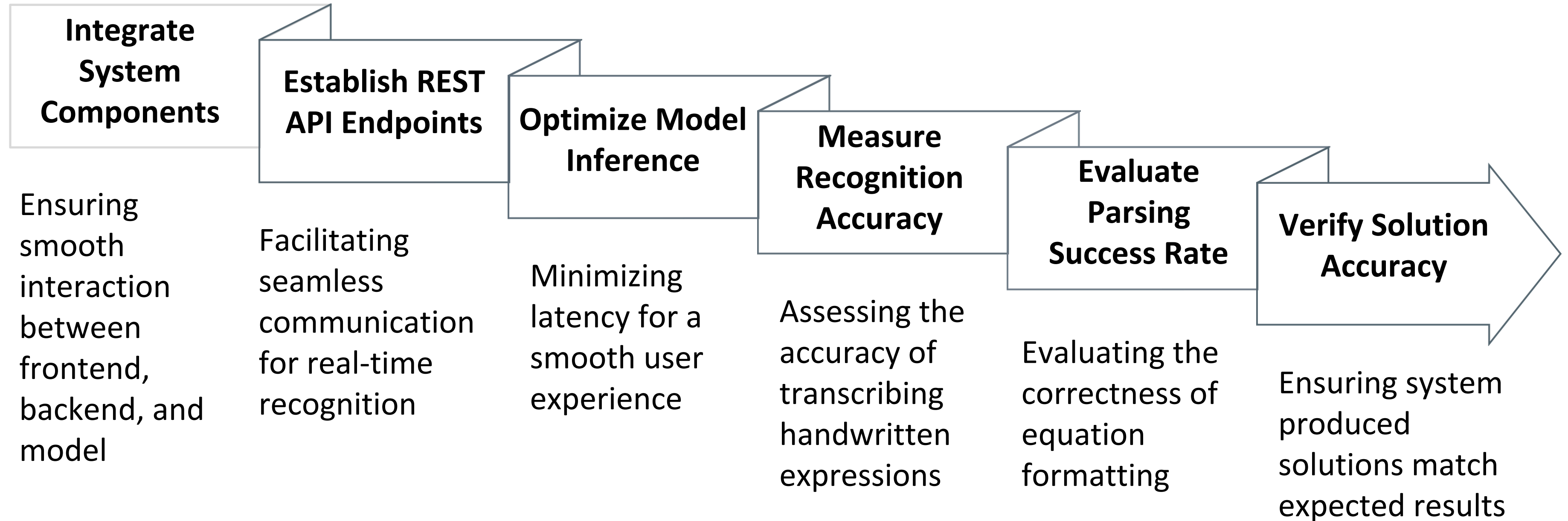
Methods to create hierarchical representations of mathematical expressions.



TrOCR Framework

A transformer-based OCR system capturing long-range dependencies and context.

Methodology: Integration and Testing



How we are different than others???

Simplicity

Easy to use with user friendly environment

0

Diverse section at single place

Integrates performances of ability to solve and graph
generation is single place

Platform for Researchers

Even without solving Arithmetic and Geometry it provides a
place for researchers not just for explorer

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